Sales Forecasting

1. Importing Libraries

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.express as px
import plotly.graph_objects as go
import warnings
warnings.filterwarnings('ignore')

from statsmodels.graphics.tsaplots import plot_acf
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.seasonal import seasonal_decompose
```

Learning Goals

- Understand the importance of sales forecasting to the firm
- Differentiate among market potential, sales potential, sales forecast, sales quotas and sales budget
- Explain the concept of market & sales potentials
- Discuss economic, industry, company & product forecast
- Outline various qualitative & quantitative methods of sales forecasting
- Discuss who is responsible for sales forecasting
- Understand how to evaluate sales forecast effectiveness

Sales forecasting is the most important planning task within any company – large or small

Importance

- Forecasting of sales provides the starting point of assumptions used in various planning activities and for the development of short term financial control system
- All functional areas of an organization have a planning task, and all their projections and future estimates depend upon the sales forecast. (Production, HR, Finance)

Sales Forecasting

1. Market Potential
2. Sales Potential
3. Market Share

Sales Forecasting

Market Potential

- Highest possible expected industry sales of a good or service in a market for a given time period.

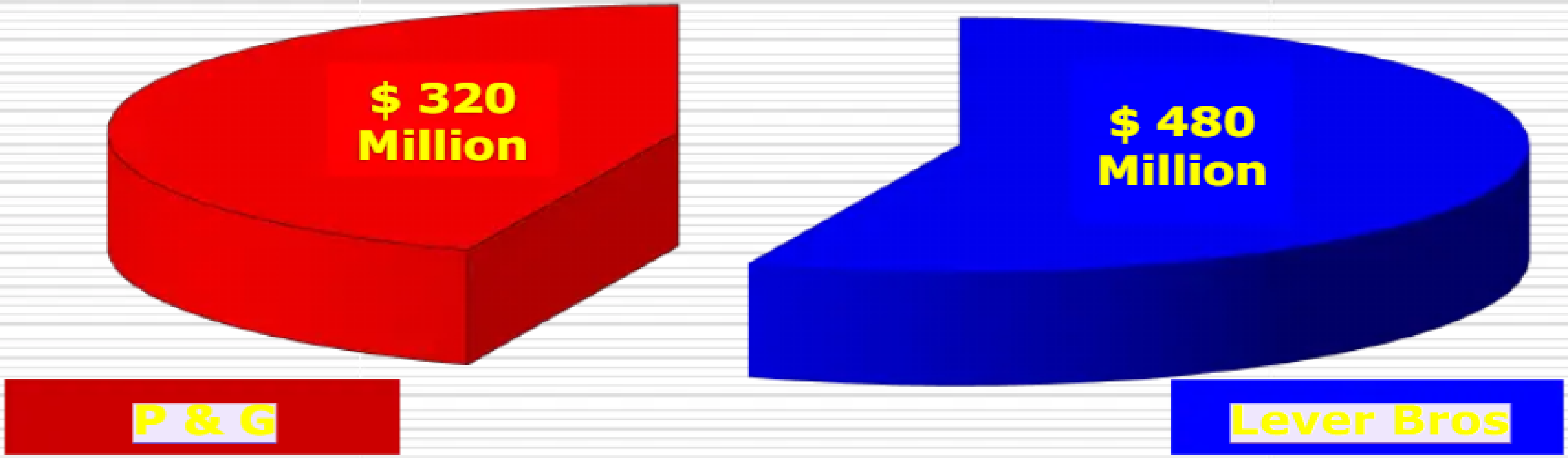
Detergents (Total Market)

$800
Million

Sales Forecasting

Sales Potential

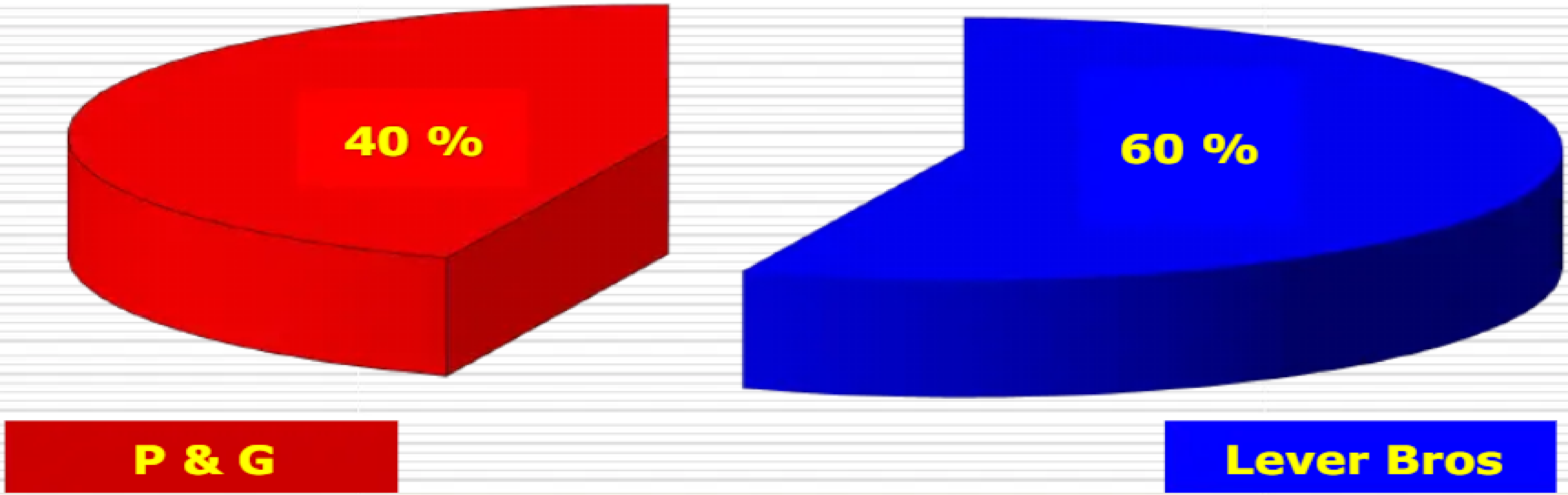
- A firm's share of the market potential



Sales Forecasting

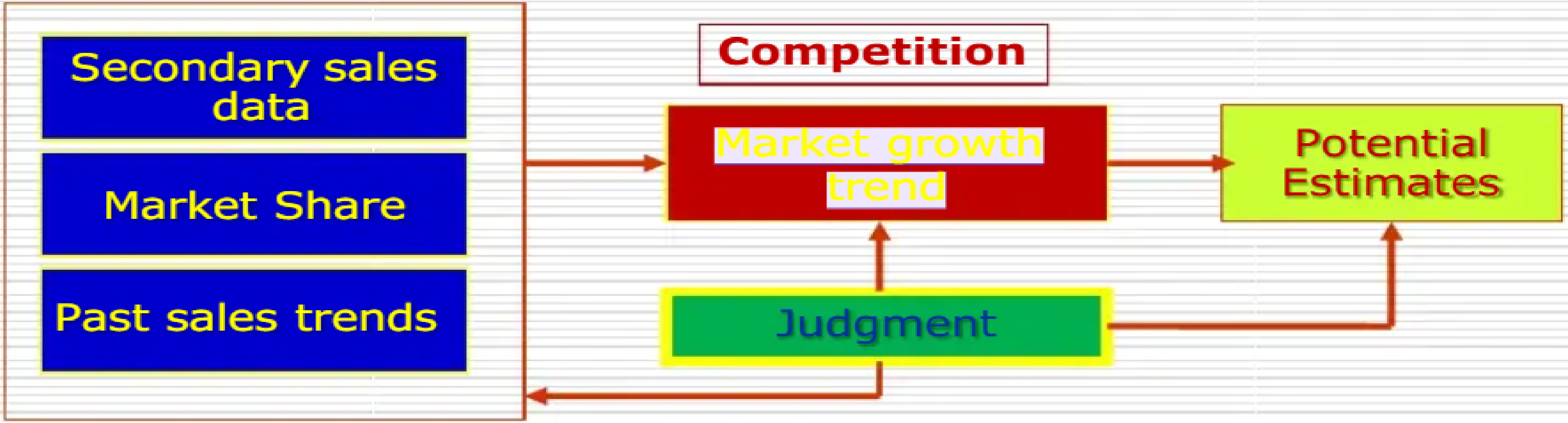
Market share

- Percentage of a market controlled by a company or product



Information Sources

Estimating Market & Sales Potential



Guidelines for Forecasting

1. Use multiple methods
2. Pick the right method for your business
3. Use as much info as you can
4. Plan for multiple scenarios
5. Track your progress and adjust the forecast

The Forecasting Process

Look at Sales Potential: Gather Internal and External Info → **Forecast Sales** → **Company-Wide Sales Forecast** → **Set Quotas for the Period**

External

- Customer & distributor surveys
- Market research conducted by other firms

Government-generated info related to economy

- Experts' opinions

Internal

- Info generated by quantitative methods using sales data
- Sales mgrs' estimate for their team
- Reps' estimates of their sales
- Executives' opinions
- Firm's plans

Forecast Sales

- By product
- By territory
- By customer type
- By time period

Set Quotas for the Period

- For individual territories
- For regions